Project Documentation: User Requirements and Analysis

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1. Introduction & Scope

1.1 Project Background

The Canadian retail market is influenced by multiple conditions, such as regional economic conditions, industry developments, e-commerce growth, and market competition. However, most of the relevant information is distributed across multiple sources and data formats, making it challenging for business decision-makers to obtain a comprehensive understanding of market trends and make decisions on whether to expand or contract their business.
Our Canadian Retail Market Intelligence Project aims to address this challenge by acquiring, preparing, and analyzing retail-related data from Statistics Canada and the Competition Bureau Canada.

1.2 Project Objectives

The primary objective of this project is to develop a data-driven analytical solution to support business decision-making in Canada's retail sector.

Specifically, the project aims to:

1. Acquire and process structured, semi-structured, and unstructured data from reliable public sources.
2. Clean, transform, validate, and integrate the collected datasets using Python and MySQL.
3. Analyze regional retail performance, industry trends, e-commerce activity, and relevant economic indicators.
4. Present analytical findings through SQL queries and Power BI visualizations to support business decisions.

1.3 Project Scope

The project scope includes:

1. Data Acquisition: Collecting structured retail sales data and semi-structured economic indicators from Statistics Canada, as well as unstructured web content from the Competition Bureau Canada.

2. Data Preparation: Using Python to clean, transform, and prepare datasets for analysis, including web scraping and text processing.
3. Database Management: Designing and implementing a relational database using MySQL to store and manage the collected datasets.
4. Data Validation and Integration: Evaluating data quality and integrating relevant information from multiple sources.
5. Data Analysis: Using SQL to examine regional retail performance, industry trends, e-commerce activity, and economic conditions.
6. Data Visualization: Developing Power BI dashboards to present market trends and support business decision-making.

The project focuses on historical retail market intelligence using publicly available Canadian data.

1.4 Stakeholders and Roles

The Canadian Retail Market Intelligence system is designed for retail professionals who require reliable market information to support business decision-making. The primary stakeholders and their responsibilities are outlined below.

Stakeholder	Role	Responsibilities
Retail Strategy Managers	Primary Users	Monitor regional and industry retail performance, evaluate market trends, and support strategic planning.
Business Analysts	Primary Users	Analyze retail sales, e-commerce trends, and economic indicators to generate actionable business insights.
Business Development Managers	Secondary Users	Identify potential market opportunities and evaluate regional expansion strategies.
Data Analysts / Database Administrators	System Administrators	Manage data acquisition, maintain the MySQL database, validate data

		quality, and ensure reliable data access.

2. Functional Requirements

The Canadian Retail Market Intelligence System functional requirements define the expected behaviours and capabilities. The system must be able to store and analyze structured, semi-structured, and unstructured Canadian retail market data, identify historical trends, and present analytical results to support business decision-making. Requirements are prioritized according to their importance to the project's core objectives.

FR-01 Store Prepared Datasets

High priority

The system must store the previously cleaned structured, semi-structured, and unstructured data from Statistics Canada and the Competition Bureau of Canada in their corresponding MySQL tables: retail_sales, economic_indicators, and web_articles.

FR-02 Validate Stored Data

High priority

The system must allow the project team to verify the completeness and consistency of the data stored in MySQL.

FR-03 Retrieve and Filter Retail Sales

High priority

The system must allow users to retrieve and filter historical retail sales records by reporting data, geography, industry, and available sales categories.

FR-04 Analyze Economic Indicators

High priority

The system must allow users to retrieve and summarize selected Canadian economic indicators by indicator, geography, and reference period.

FR-05 Analyze Retail Competition Information

Medium priority

The system must allow users to examine the collected Competition Bureau article data using publication metadata and keyword metrics related to competition, prices, groceries, consumers, and online retail.

FR-06 Generate and Present Retail Market Trends

High priority

The system must generate summaries of historical retail sales data and present selected analytical results through Power BI visualization to help users identify market trends and support business decision-making.

3. Non-Functional Requirements

The Canadian Retail Market Intelligence system shall meet the following non-functional requirements. These requirements describe the quality standards the system must satisfy while storing, processing, and analyzing retail data from Statistics Canada and the Competition Bureau Canada.

NFR-01 Data Integrity

Primary keys and appropriate uniqueness constraints shall be used to maintain record integrity and prevent unintended duplicate records. Relationships between datasets shall be established where compatible fields and valid relationships exist.

NFR-02 Data Quality

All data collected for the project shall be checked before it is loaded into the database. The system shall identify missing values, invalid dates, and incorrect data types, and these issues shall be corrected during the cleaning and transformation stages in Python. This ensures that the analysis and the Power BI dashboards are based on complete and valid data.

NFR-03 Security

Access to the database shall be restricted to authorized team members. Sensitive credentials, such as database usernames and passwords, shall be protected and shall not be stored in shared scripts or documents. This reduces the risk of unauthorized access to the project data.

NFR-04 Performance

The SQL queries used to analyze regional retail performance, industry trends, and

e-commerce activity shall execute efficiently. Indexes shall be applied to frequently searched fields, such as region, industry, and date, so that results are returned within a reasonable time as the dataset grows.

NFR-05 Reliability

The system shall import and retrieve datasets consistently. Running the same import process or query multiple times shall produce the same results, and failed or incomplete imports shall be identified and corrected. This allows users to trust the results shown in the SQL outputs and dashboards.

NFR-06 Maintainability

The SQL and Python scripts shall be organized, clearly named, and commented so that any team member can understand and update them. The database structure, including tables, fields, and relationships, shall be documented. This makes the system easier to maintain and extend after the initial development.

NFR-07 Compliance

The project shall use only publicly available data and shall use it responsibly. The team shall follow the usage requirements of Statistics Canada and the Competition Bureau Canada, including proper citation of data sources. Web scraping activities shall respect the terms of use of the source websites.

4. Acceptance Criteria

4.1 Functional Requirements

The following acceptance criteria define measurable and testable conditions for verifying that each functional requirement has been met.

4.1.1 Store Prepared Datasets

1. The structured retail sales CSV is mapped to the retail_sales table.
2. The semi-structured economic indicators CSV is mapped to the economic_indicators table.
3. The unstructured article CSV is mapped to the web_articles table.
4. The data is mapped to the corresponding database columns and data types.

5. SQL SELECT queries successfully retrieve records from all three MySQL tables: retail_sales, economic_indicators, and web_articles.

4.1.2 Validate Stored Data

1. The number of records imported into each MySQL table matches the number of valid records in its corresponding cleaned CSV file.
2. Import record counts can be compared with the corresponding cleaned CSV files.
3. The system's primary keys and uniqueness constraints prevent duplicate records from disrupting the defined rules.
4. SQL queries identify missing required values, invalid formats, and potential data inconsistencies.
5. Discrepancies identified during validation are investigated and documented.

4.1.3 Retrieve and Filter Retail Sales

1. SQL queries can filter retail sales records by date, geography, and industry.
2. Users can retrieve sales values for selected regions and reporting periods.
3. Results include the relevant sales measures and reporting attributes.
4. Query results match the corresponding records stored in the database.

4.1.4 Analyze Economic Indicators

1. SQL queries can retrieve records for selected economic indicators.
2. Numeric values can be used to calculate summaries and comparisons.
3. Users can compare available indicator values across reporting periods or geographical locations.
4. Results identify the indicator and reference period being analyzed.
5. Comparisons with retail sales are performed only when the relevant geographic and temporal attributes are compatible.

4.1.5 Analyze Retail Competition Information

1. SQL queries can retrieve articles by publication date, year, and source.
2. Users can retrieve keyword counts and frequencies per 1000 words.
3. Results allow comparison of keyword metrics across the available articles.
4. Each article can be identified through its title and source.
5. Results are interpreted as indicators of article content rather than direct measurements of market competition.

4.1.6 Generate and Present Retail Market Trends

1. SQL queries calculate relevant sales summaries, such as totals or averages.
2. Analytical results can be made available to Power BI through a connection or export method.

3. Power BI visualization presents selected trends based on the available datasets.
4. Values displayed in the visualizations are consistent with the validated data and query results.

4.2 Non-Functional Requirements Acceptance Criteria

The following criteria are used to check whether the system meets the non-functional requirements listed in Section 3.

4.2.1 Data Integrity

1. Each table has a primary key, and duplicate primary key values are not allowed.
2. Uniqueness constraints are checked to ensure that duplicate records do not violate the database rules.
3. Data from different sources is only combined when the relevant fields are compatible.

4.2.2 Data Quality

1. The cleaned datasets are checked for missing values, invalid dates, and incorrect data types before being imported into MySQL.
2. Any data quality issues found during validation are corrected or documented.
3. The data stored in MySQL matches the cleaned datasets used for import.

4.2.3 Security

1. Database access is limited to authorized team members.
2. Database passwords and other sensitive credentials are not included in the shared GitHub repository or project documents.

4.2.4 Performance

1. SQL queries can retrieve and summarize retail sales data without errors.
2. Indexes are available on commonly used fields, such as date, geography, and industry.
3. The execution time of the main analytical queries is recorded to check whether the queries run efficiently.

4.2.5 Reliability

1. Running the same SQL query on unchanged data produces the same results.
2. The number of imported records is checked against the cleaned CSV files to identify incomplete imports.
3. Any failed imports can be identified and corrected before the data is used for analysis.

4.2.6 Maintainability

1. Python and SQL scripts are organized in clearly named files and folders.

2. The database tables and their main fields are documented so that other team members can understand the structure.
3. Scripts include comments where needed to explain important processing steps.

4.2.7 Compliance

1. The sources of the datasets are identified and documented in the project.
2. The project uses publicly available data from Statistics Canada and the Competition Bureau Canada.
3. The web scraping process follows the applicable usage requirements of the source websites.

5. Revision History

Version	Date	Author	Description of Changes	Status
1.0	10/21	Quan Yuan	Create a shared document and draft the document	Draft
1.1	10/21	Quan Yuan	Complete part one: Introduction & scope	Draft
1.2	10/22	Jiya Khadgi	Complete part three: Non-Functional Requirements	Draft
1.3	10/22	Santiago Pinzon	Complete part two and half part for: Functional Requirements and Acceptance Criteria	Draft
1.4	10/23	Quan Yuan	Complete Part 4: Non-Functional Requirements and AC	Draft

2.0	10/23	Quan Yuan	Review the entire text and make corrections to grammar, formatting, and content.	Approved